PhotolithoTwin

Simulation Report

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PhotolithoTwin · Digital Twin Photolithography

Generated: 14 October 2026 · Process step: Final Pattern

OPTICAL SYSTEM

Wavelength	248nm (DUV)
Numerical Aperture (NA)	0.63
Partial Coherence (Ã)	0.40
Defocus Offset	0 nm
Resolution Limit (k•·»/NA)	240 nm
Depth of Focus (»/NA²)	625 nm
PSF Blur Ã	161 nm
Image Log Slope (ILS)	11.16 /µm

PHOTORESIST

Resist Type	Negative (SU-8 / cross-linking)
Resist Thickness	500 nm
Contrast (³)	7.0
Dill C (absorptivity)	0.040 cm²/mJ

EXPOSURE & BAKE

Exposure Dose	120 mJ/cm²
Soft Bake	95°C · 60s
Post-Exposure Bake	110°C · 60s
Hard Bake	120°C
Development Time	45s
Diffusion Strength	0.00

CRITICAL DIMENSION (CD) METRICS

CD…€ (linewidth at 50% height)	2500 nm
CDˆ€ (linewidth at 80% height)	2250 nm
CD Bias (CD…€" target)	-500 nm
Sidewall Angle	40.7°
Normalized ILS (NILS)	2.679

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